

Unit 9

Orthonormal Bases

Unit Goals

- Be able to define the meaning of an orthonormal basis for a vector space.
- Be able to explain why projections are simplified when using an orthonormal basis.
- Be able to find an orthonormal basis for a vector space using the Gram-Schmidt algorithm.
- Be able to express the results of Gram-Schmidt orthonormalization in a QR decomposition.

Unit 9.1

Orthonormal bases

Bases for a vector space

Definitions:

- Recall that a **basis** for a vector space is a linearly independent spanning set.
- ANY linear independent spanning set is a basis.
 - For $\mathbb{R}^2$, $\left\{ \begin{bmatrix} 15 \\ 15 \end{bmatrix}, \begin{bmatrix} 15.1 \\ 14.9 \end{bmatrix} \right\}$, $\left\{ \begin{bmatrix} 10 \\ 8 \end{bmatrix}, \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix} \right\}$, and $\left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \right\}$ are all bases.
 - They look very different when we sketch them!
- Many things become much simpler when our basis vectors are orthogonal, and even more so when they all have a length of 1.

Orthogonal and orthonormal bases

Definitions:

- An **orthogonal basis** is one where all the basis vectors are orthogonal to one another:

$$\vec{v}_i^T \vec{v}_j = 0, \quad \forall i \neq j$$

- An **orthonormal basis** is one where all the basis vectors are orthogonal to one another AND they all have length (norm) of one:

$$\vec{v}_i^T \vec{v}_j = 0, \forall i \neq j, \quad \vec{v}_i^T \vec{v}_i = 1, \forall i$$

Projections with an orthogonal basis.

- Projections and pseudoinverses require a matrix inverse:

$$\text{proj}_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}, \quad A^+ = (A^T A)^{-1} A^T$$

- The columns of A are a basis for V :

$$A = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n]$$

- The entries of $A^T A$ are the dot products of the basis vectors:

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & \vec{v}_1^T \vec{v}_2 & \cdots & \vec{v}_1^T \vec{v}_n \\ \vec{v}_2^T \vec{v}_1 & \vec{v}_2^T \vec{v}_2 & \cdots & \vec{v}_2^T \vec{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{v}_n^T \vec{v}_1 & \vec{v}_n^T \vec{v}_2 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix}$$

- If the basis is orthogonal, $A^T A$ is a diagonal matrix; matrix inverse is easy!

Projections with an orthogonal basis.

$$\text{proj}_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}$$

$$A = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n], \quad \vec{v}_i^T \vec{v}_j = 0, \forall i \neq j$$

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & 0 & \cdots & 0 \\ 0 & \vec{v}_2^T \vec{v}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix} \rightarrow (A^T A)^{-1} = \begin{bmatrix} \frac{1}{\vec{v}_1^T \vec{v}_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\vec{v}_2^T \vec{v}_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\vec{v}_n^T \vec{v}_n} \end{bmatrix}$$

Projections with an orthogonal basis.

$$\text{proj}_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}$$

$$(A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} \frac{1}{\vec{v}_1^T \vec{v}_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\vec{v}_2^T \vec{v}_2} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \frac{1}{\vec{v}_n^T \vec{v}_n} \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \\ \vdots \\ \vec{v}_n^T \end{bmatrix} \vec{u} = \begin{bmatrix} \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \\ \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \\ \vdots \\ \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \end{bmatrix}$$

Projections with an orthogonal basis.

$$\text{proj}_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}$$

$$A(A^T A)^{-1} A^T \vec{u} = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \begin{bmatrix} \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \\ \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \\ \vdots \\ \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \end{bmatrix} = \frac{\vec{v}_1^T \vec{u}}{\vec{v}_1^T \vec{v}_1} \vec{v}_1 + \frac{\vec{v}_2^T \vec{u}}{\vec{v}_2^T \vec{v}_2} \vec{v}_2 + \frac{\vec{v}_n^T \vec{u}}{\vec{v}_n^T \vec{v}_n} \vec{v}_n$$

- When we have an orthogonal basis, the projection onto the vector space is simply the **sum of the projections onto each basis vector!**

Projections with an orthonormal basis.

- For an orthonormal basis:

$$\vec{v}_i^T \vec{v}_j = 0, \forall i \neq j, \quad \vec{v}_i^T \vec{v}_i = 1, \forall i$$

- $A^T A$ is an identity matrix!:

$$A^T A = \begin{bmatrix} \vec{v}_1^T \vec{v}_1 & \vec{v}_1^T \vec{v}_2 & \cdots & \vec{v}_1^T \vec{v}_n \\ \vec{v}_2^T \vec{v}_1 & \vec{v}_2^T \vec{v}_2 & \cdots & \vec{v}_2^T \vec{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{v}_n^T \vec{v}_1 & \vec{v}_n^T \vec{v}_2 & \cdots & \vec{v}_n^T \vec{v}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I$$

- Projections and pseudoinverses are greatly simplified:

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u} \rightarrow \textcolor{brown}{proj}_V(\vec{u}) = A A^T \vec{u}$$

$$A^+ = (A^T A)^{-1} A^T \rightarrow \textcolor{brown}{A}^+ = A^T$$

Projections with an orthogonal basis.

$$\text{proj}_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u} = AA^T \vec{u}$$

$$AA^T \vec{u} = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n] \begin{bmatrix} \vec{v}_1^T \vec{u} \\ \vec{v}_2^T \vec{u} \\ \vdots \\ \vec{v}_n^T \vec{u} \end{bmatrix} = (\vec{v}_1^T \vec{u}) \vec{v}_1 + (\vec{v}_2^T \vec{u}) \vec{v}_2 + (\vec{v}_n^T \vec{u}) \vec{v}_n$$

- When we have an orthonormal basis, the projection onto the vector space is simply the sum of the projections onto each basis vector, and each projection is just the vector scaled by a single dot product.

Unit 9.2

Gram-Schmidt Orthonormalization

How to find an orthonormal basis?

The Gram-Schmidt algorithm:

- A robust method to generate an orthonormal basis starting from any basis.

Key ideas:

- The projection can be used express a vector into a parallel and orthogonal component, relative to some vector space.
- Projections onto vector space are easy to evaluate when we are working with an orthogonal basis set; they are sums of projections onto of individual vectors.
- Dividing by the length of a vector yields a unit vector in the same direction.

Gram-Schmidt

Goal: Starting with any basis for a vector space, $\{\vec{v}_i\}$ and create a new basis $\{\vec{w}_i\}$ that is orthogonal!

Method:

- The first vector of the new basis is just the first vector of the original basis; this vector spans a line in the original vector space:

$$\vec{w}_1 = \vec{v}_1, \quad W_1 = \text{span}(\{\vec{w}_1\})$$

- We construct a vector that is orthogonal to this line by finding the orthogonal component of the second vector from the original set; we now have an orthogonal pair of vector that spans a plane in the original vector space:

$$\vec{w}_2 = \vec{v}_2 - \text{proj}_{W_1}(\vec{v}_2) = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1, \quad W_2 = \text{span}(\{\vec{w}_1, \vec{w}_2\})$$

Gram-Schmidt

Method, continued:

- We construct a vector that is orthogonal to this plane by finding the orthogonal component of the third vector from the original set; we now have an orthogonal triplet of vector that spans a 3D subspace of the original vector space:

$$\vec{w}_3 = \vec{v}_3 - \text{proj}_{W_2}(\vec{v}_3) = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2, \quad W_3 = \text{span}(\{\vec{w}_1, \vec{w}_2, \vec{w}_3\})$$

- We continue as needed, until we reach the dimension of the space. At each step we add another orthogonal component:

$$\vec{w}_i = \vec{v}_i - \text{proj}_{W_{i-1}}(\vec{v}_i) = \vec{v}_i - \frac{\vec{w}_1^T \vec{v}_i}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_i}{\vec{w}_2^T \vec{w}_2} \vec{w}_2 - \cdots - \frac{\vec{w}_{i-1}^T \vec{v}_i}{\vec{w}_{i-1}^T \vec{w}_{i-1}} \vec{w}_{i-1}$$

Gram-Schmidt

Normalization

- We now have an orthogonal basis:

$$V = \text{span}(\{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}), \quad \vec{w}_i^T \vec{w}_j = 0, \forall i \neq j$$

- We obtain an orthonormal basis by dividing each basis vector by its length:

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|} \right) \vec{w}_i, \forall i$$

$$V = \text{span}(\{\vec{q}_1, \vec{q}_2, \dots, \vec{q}_n\}), \quad \vec{q}_i^T \vec{q}_j = 0, \forall i \neq j, \quad \|\vec{q}_i\| = 1, \forall i$$

Gram-Schmidt

An example: $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = \text{span}(S)$

$$\vec{w}_1 = \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix} - \left(\frac{10}{5} \right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}$$

Gram-Schmidt

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = \text{span}(S) \quad W_2 = \text{span} \left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix} \right\} \right)$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2 = \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} - \left(\frac{10}{5} \right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \left(\frac{8}{6} \right) \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 1 \\ -2/3 \\ 1 \\ -2/3 \end{pmatrix}$$

Gram-Schmidt

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 4 \\ 3 \\ 0 \end{pmatrix} \right\}, V = \text{span}(S) \quad W_3 = \text{span} \left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -2/3 \\ 1 \\ -2/3 \\ 1 \\ -2/3 \end{pmatrix} \right\} \right)$$

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|} \right) \vec{w}_i \rightarrow \vec{q}_1 = \left(\frac{1}{\sqrt{5}} \right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \vec{q}_2 = \left(\frac{1}{\sqrt{6}} \right) \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \\ -1 \end{pmatrix}, \vec{q}_3 = \left(\frac{1}{\sqrt{30}} \right) \begin{pmatrix} -2 \\ 3 \\ -2 \\ 3 \\ -2 \end{pmatrix}$$

Unit 9.2.1

**Worked examples with
orthonormal bases**

Projections with orthonormal bases.

- Find the projection of the vector $\vec{u}$ onto the subspace of spanned by $\{\vec{q}_1, \vec{q}_2, \vec{q}_3, \vec{q}_4\}$ and, where:

$$\vec{u} = (1, 2, 3, 4, 5)$$

$$\vec{q}_1 = \left(\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2} \right), \vec{q}_2 = (0, 0, 1, 0, 0),$$

$$\vec{q}_3 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0, 0, 0 \right), \vec{q}_4 = \left(0, 0, 0, \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right)$$

Projections with orthonormal bases.

$$Q = [\vec{q}_1 \quad \vec{q}_2 \quad \vec{q}_3 \quad \vec{q}_4]$$

$$\text{proj}_V(\vec{u}) = Q(Q^T Q)^{-1} Q^T \vec{u}$$

$$Q = \begin{bmatrix} 1/2 & 0 & 1/\sqrt{2} & 0 \\ 1/2 & 0 & -1/\sqrt{2} & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 0 & 1/\sqrt{2} \\ 1/2 & 0 & 0 & -1/\sqrt{2} \end{bmatrix} \rightarrow Q^T Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \text{proj}_V(\vec{u}) = Q Q^T \vec{u}$$

Projections with orthonormal bases.

$$\text{proj}_V(\vec{u}) = QQ^T\vec{u}$$

$$Q^T\vec{u} = \begin{bmatrix} 1/2 & 1/2 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1 & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

Projections with orthonormal bases.

$$\text{proj}_V(\vec{u}) = QQ^T\vec{u} = \begin{bmatrix} 1/2 & 0 & 1/\sqrt{2} & 0 \\ 1/2 & 0 & -1/\sqrt{2} & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 0 & 1/\sqrt{2} \\ 1/2 & 0 & 0 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 5/2 \\ 7/2 \\ 3 \\ 5/2 \\ 7/2 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- Find an orthonormal basis for the vector space:

$$V = \text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- First generate an orthogonal basis:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$\vec{w}_1 = \vec{v}_1$$

$$\vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2$$

Gram-Schmidt orthonormalization.

- First generate an orthogonal basis:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$\vec{w}_1 = \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{w}_1^T \vec{w}_1 = 4, \vec{w}_1^T \vec{v}_2 = 6,$$

$$\vec{w}_2 = \vec{v}_2 - \frac{\vec{w}_1^T \vec{v}_2}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{6}{4}\right) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- First generate an orthogonal basis:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad \vec{w}_1^T \vec{w}_1 = 4, \vec{w}_1^T \vec{v}_3 = 6, \quad \vec{w}_2^T \vec{w}_2 = 4, \vec{w}_2^T \vec{v}_3 = 0$$

$$\vec{w}_3 = \vec{v}_3 - \frac{\vec{w}_1^T \vec{v}_3}{\vec{w}_1^T \vec{w}_1} \vec{w}_1 - \frac{\vec{w}_2^T \vec{v}_3}{\vec{w}_2^T \vec{w}_2} \vec{w}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix} - \left(\frac{6}{4}\right) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{0}{4}\right) \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- Then normalize:

$$\vec{q}_i = \left(\frac{1}{\|\vec{w}_i\|} \right) \vec{w}_i, \quad \|\vec{w}_i\| = \sqrt{\vec{w}_i^T \vec{w}_i}$$

$$\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \vec{w}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\vec{w}_1^T \vec{w}_1 = 4, \quad \vec{w}_2^T \vec{w}_2 = 4, \quad \vec{w}_3^T \vec{w}_3 = 4$$

$$\vec{q}_1 = \left(\frac{1}{2} \right) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \vec{q}_2 = \left(\frac{1}{2} \right) \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \vec{q}_3 = \left(\frac{1}{2} \right) \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$

Gram-Schmidt orthonormalization.

- Our orthonormal basis is:

$$Q = \{\vec{q}_1, \vec{q}_2, \vec{q}_3\}, \quad V = \text{span}(Q)$$

$$\vec{q}_1 = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \quad \vec{q}_2 = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \quad \vec{q}_3 = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}$$